Cofunction identities

1

a 56° b 28° c $\frac{\pi}{3}$ d $\frac{3\pi}{14}$

2

a $\sin 53^\circ$ b $\cos 14^\circ$ c $\cot 22^\circ$ d $\tan 47^\circ$

e $\sec 55^\circ$ f $\sin \frac{7\pi}{18}$ g $\cos \frac{2\pi}{5}$ h $\tan \frac{7\pi}{18}$

i $\csc \frac{7\pi}{18}$ j $\sin \left(\frac{5\pi - 2}{10}\right)$

3

a

i $\frac{a}{c}$ ii $\frac{a}{c}$ iii $\frac{b}{c}$ iv $\frac{b}{c}$

b If two angles are complementary then the sine of one angle will equal the cosine of the other angle.

c 0.32

d 90°

4

a $\cos p$ b $\tan x$ c $\frac{1}{7}$ d 1

e $\sin y$

5 0.19

6 0.46

7

a No solution provided. b No solution provided.

Trigonometric equations

8

a $\theta = 65^\circ$ b $\theta = 5^\circ$ c $\theta = \frac{3\pi}{8}$ d $\theta = \frac{\pi}{10}$

9 $c=\frac{5\pi}{36}$

10 $c=5$

11 $B=14$

12 $A=4896$

13

a $x=5^{\circ}$

b $x=\frac{\pi}{180}$